Infinitude of OEIS A131409

Consecutive Semiprimes with Common Prime Factor 3

Primary author (AI): ChatGPT (OpenAI)

11 November 2026

Abstract

For every integer $r \geq 2$ and every prime p, the shared prime-pattern construction produces infinitely many blocks of r consecutive squarefree semiprimes whose pairwise greatest common divisors equal p. The blocks arise from fixed prime offsets with successively increasing, divisible gaps. The construction uses quantitative prime-cluster theory, congruences, and the sparsity of prime powers. Taking $r = 3$ and $p = 3$ proves that OEIS A131409, which records middle terms of consecutive semiprime triples with common factor 3, has infinitely many terms. This is an unconditional deduction from established Maynard–Tao theory; publication priority is not claimed.

1 Notation and a prime-pattern theorem

A *semiprime* is a positive integer having exactly two prime factors counted with multiplicity; prime squares are included. A *squarefree semiprime* is the product of two distinct primes. Numbers $s_1 < \cdots < s_r$ are *consecutive semiprimes* if each is a semiprime and no other semiprime lies strictly between any adjacent pair. Consecutiveness refers to the ordered sequence of semiprimes, not to consecutive integers.

For a positive integer n, let $\omega(n)$ denote the number of distinct prime divisors of n, with $\omega(1) = 0$. A finite family of integer linear forms is *admissible* if, for every prime p, some integer argument makes every form nonzero modulo p.

Theorem 1 (Prime-pattern theorem)**.** *For every integer $m \geq 2$, there exist fixed integers*

$$0 = h_1 < h_2 < \cdots < h_m$$

with the following property. For every finite set $A \subseteq \{2, 3, 4, \ldots\}$, there are infinitely many positive integers q such that:

1. *$q + h_1, q + h_2, \ldots, q + h_m$ are consecutive primes;*
2. *writing $\delta_i = h_{i+1} - h_i$, we have*

$$\delta_i < \delta_{i+1} \quad \text{and} \quad \delta_i \mid \delta_{i+1} \qquad (1 \leq i \leq m - 2);$$

3. *for every $a \in A$, every $1 \leq i \leq m - 1$, and every integer x,*

$$a(q + h_i) < x < a(q + h_{i+1}) \quad \Longrightarrow \quad \omega(x) \geq 3.$$

The offsets $h_1, \ldots, h_m$ depend only on m, not on A.

Established input (Quantitative Maynard–Tao theorem). *For each integer $m \geq 2$, there is an integer $k \geq m$ such that, for every fixed admissible family of distinct linear forms*

$$L_j(n) = u_j n + v_j, \qquad u_j \in \mathbb{N}, \quad v_j \in \mathbb{Z}_{\geq 0}, \quad 1 \leq j \leq k,$$

there are constants $c > 0$ and X_0, depending on the family, such that for every $X \geq X_0$,

$$\#\{n \in \mathbb{Z} \cap [X, 2X] : \text{at least } m \text{ of the } L_j(n) \text{ are prime}\} \geq \frac{cX}{(\log X)^k}.$$

This fixed-coefficient form follows from Maynard's Theorem 3.4 in [1], taking $y = X$ and, for example, $\varepsilon = 1/10$. All coefficient bounds in that theorem hold for sufficiently large X because the forms here are fixed.

Proof of Theorem 1. Fix $m \geq 2$, and choose k from the established input. Put

$$Q = \prod_{\substack{p \leq k \\ p \text{ prime}}} p, \qquad b_1 = 0,$$

and define, for $1 \leq j < k$,

$$b_{j+1} = b_j + Q(b_j + 1) \prod_{1 \leq u < v \leq j} (b_v - b_u),$$

where an empty product equals 1. Write $H = \{b_1, \ldots, b_k\}$.

1. Admissibility and gap divisibility. The sequence $b_1, \ldots, b_k$ is strictly increasing, and each b_j is divisible by Q. For a prime $p \leq k$, every form $T + b_j$ is nonzero modulo p when $T \equiv 1 \pmod{p}$. For a prime $p > k$, the k forms exclude at most $k < p$ residue classes of T modulo p. Thus $\{T + b_j : 1 \leq j \leq k\}$ is admissible.

If $i < j < \ell$, then every increment $b_{r+1} - b_r$ with $j \leq r < \ell$ contains the factor $b_j - b_i$. Consequently,

$$b_j - b_i \mid b_\ell - b_j.$$

Also, since $Q \geq 2$ and every factor in the product is positive,

$$b_\ell - b_j \geq b_{j+1} - b_j \geq Q(b_j + 1) > b_j \geq b_j - b_i.$$

It follows that the adjacent gaps in every increasing subsequence of H strictly increase and each divides the next.

2. Congruences for a fixed finite multiplier set. Fix a finite $A \subseteq \{2, 3, \ldots\}$ and put

$$B = A \cup \{1\}, \qquad M = \prod_{a \in A} a.$$

Here $M = 1$ if A is empty. For each $a \in B$, define

$$E_a = \{e \in \mathbb{Z} : 0 \leq e \leq ab_k, \ e \notin \{ab_1, \ldots, ab_k\}\}.$$

For each pair (a, e) with $a \in B$ and $e \in E_a$, choose three primes $\ell_{a,e,1}$, $\ell_{a,e,2}$, $\ell_{a,e,3}$, with all these primes distinct, so large that each satisfies

$$\ell_{a,e,t} > \max B, \qquad \ell_{a,e,t} > \max_{1 \leq j \leq k} |e - ab_j|.$$

There are only finitely many pairs, so such choices are possible. Impose the congruences

$$T \equiv -ea^{-1} \pmod{\ell_{a,e,t}} \qquad (t = 1, 2, 3),$$

where the inverse exists because $\ell_{a,e,t} > a$. Then each of these three primes divides $aT + e$. None divides any candidate $T + b_j$, because

$$T + b_j \equiv (ab_j - e)a^{-1} \not\equiv 0 \pmod{\ell_{a,e,t}}.$$

The last assertion follows from $e \neq ab_j$ and $0 < |ab_j - e| < \ell_{a,e,t}$.

For each prime $p \mid M$, also choose a residue c_p satisfying

$$c_p + b_j \not\equiv 0 \pmod{p} \qquad (1 \le j \le k),$$

which is possible by the admissibility proved above, and impose $T \equiv c_p \pmod{p}$. The auxiliary primes exceed every element of A, so none divides M. All the specified moduli are therefore distinct primes. The Chinese remainder theorem gives a residue class

$$T = Wn + r, \qquad W \ge 1, \qquad 0 \le r < W,$$

that satisfies every congruence, where W is the product of the specified primes. In particular,

$$\gcd(T + b_j, a) = 1 \qquad (a \in A,\ 1 \le j \le k).$$

For all sufficiently large n, we also have

$$\omega(aT + e) \ge 3 \qquad (a \in B,\ e \in E_a).$$

3. Many prime candidates and removal of prime powers. Consider the fixed linear forms

$$L_j(n) = Wn + r + b_j \qquad (1 \le j \le k).$$

They are admissible. Indeed, if $p \mid W$, every $L_j(n)$ is nonzero modulo p by construction. If $p \nmid W$, multiplication by W is invertible modulo p, and admissibility follows from that of the original shifts $T + b_j$.

The established input therefore gives at least $cX/(\log X)^k$ integers $n \in [X, 2X]$ for which at least m of the candidate values are prime.

Discard any n for which some $L_j(n)$ is a prime power p^ν with $\nu \ge 2$. For fixed W, r, H, all candidate values in this range are at most CX for a fixed constant C. The number of integer powers $z^\nu \le CX$ with $z \ge 2$ and $\nu \ge 2$ is

$$\le \sum_{2 \le \nu \le \log_2(CX)} (CX)^{1/\nu} = O(\sqrt{X} \log X).$$

Each L_j is injective, so the number of discarded n is $O_{W,r,H}(\sqrt{X} \log X)$. Since

$$\sqrt{X} \log X = o\left(\frac{X}{(\log X)^k}\right),$$

infinitely many of the prime-producing n remain.

For such an n, put $T = Wn + r$, and list the first m prime candidate values in increasing order as

$$P_i = T + b_{j_i}, \qquad j_1 < \cdots < j_m.$$

They are consecutive primes: every integer between adjacent P_i, P_{i+1} is either a nonprime candidate, or has an offset in E_1 and hence at least three distinct prime divisors.

Now fix $a \in A$ and an integer x with $aP_i < x < aP_{i+1}$. Write $x = aT + e$. If $e \notin \{ab_1, \ldots, ab_k\}$, then $e \in E_a$, so $\omega(x) \geq 3$ by construction. Otherwise $e = ab_j$ for some $j_i < j < j_{i+1}$, and

$$x = a(T + b_j).$$

The integer $T + b_j$ is composite because it lies strictly between the consecutive primes P_i and P_{i+1}. It is not a prime power, by the discarded-case condition, so $\omega(T + b_j) \geq 2$. Its prime divisors are disjoint from those of a, by the coprimality condition. Thus

$$\omega(x) = \omega(a) + \omega(T + b_j) \geq 1 + 2 = 3.$$

This proves that, for every fixed finite A, infinitely many T give some m-element subset of H with all the required prime and dilation properties.

4. One pattern for every finite multiplier set. For each integer $L \geq 2$, let $\mathcal{P}_L$ be the collection of index sets $J = \{j_1 < \cdots < j_m\} \subseteq \{1, \ldots, k\}$ for which infinitely many positive integers T satisfy both of the following conditions:

- $T + b_{j_1}, \ldots, T + b_{j_m}$ are consecutive primes;
- for every $2 \leq a \leq L$ and every $1 \leq i < m$, every integer strictly between $a(T + b_{j_i})$ and $a(T + b_{j_{i+1}})$ has at least three distinct prime divisors.

Step 3 and the finite pigeonhole principle show that every $\mathcal{P}_L$ is nonempty. These collections are finite and

$$\mathcal{P}_2 \supseteq \mathcal{P}_3 \supseteq \mathcal{P}_4 \supseteq \cdots.$$

Hence their intersection is nonempty. Choose $J = \{j_1 < \cdots < j_m\}$ in that intersection, and put

$$h_i = b_{j_i} - b_{j_1} \qquad (1 \leq i \leq m).$$

Step 1 gives the required strict inequalities and divisibility relations for their adjacent gaps.

For any finite $A \subseteq \{2, 3, \ldots\}$, choose $L \geq 2$ such that $A \subseteq \{2, \ldots, L\}$. Since $J \in \mathcal{P}_L$, there are infinitely many suitable T. Setting $q = T + b_{j_1}$ gives $q + h_i = T + b_{j_i}$, and establishes all assertions of the theorem. □

Remark (quantifiers). The quantifiers are

$$\forall m \geq 2 \quad \exists H_m \quad \forall A \text{ finite} \quad \exists \text{ infinitely many } q.$$

The set of suitable q may depend on A. The claim concerns each interval between adjacent dilated endpoints; it does not impose a condition at the endpoints themselves.

2 Consecutive semiprime blocks

Theorem 2 (Common prime factor)**.** *For every integer $r \geq 2$, there are fixed integers*

$$0 = h_1 < h_2 < \cdots < h_r$$

such that the following holds. For every finite set P of primes, there are infinitely many positive integers q such that, for every $p \in P$, the numbers $s_i = p(q + h_i)$, $1 \leq i \leq r$, are consecutive squarefree semiprimes and satisfy $\gcd(s_i, s_j) = p$ for $1 \leq i < j \leq r$. The numbers $q + h_1, \ldots, q + h_r$ are consecutive primes. Writing $\Delta_i = s_{i+1} - s_i$, we also have

$$\Delta_i < \Delta_{i+1}, \qquad \Delta_i \mid \Delta_{i+1} \quad (1 \leq i \leq r - 2).$$

In particular, for every prime p, infinitely many such blocks exist, with bounded diameter $s_r - s_1 = ph_r$.

Proof. Take the offsets from Theorem 1 with $m = r$, and apply that theorem with $A = P$. Discard the finitely many q satisfying $q \leq \max(P \cup \{2\})$. For each remaining q, put $t_i = q + h_i$. These are distinct consecutive primes and $t_i > p$ for every $p \in P$. Thus $s_i = pt_i$ is the product of two distinct primes, so it is a squarefree semiprime.

Every integer strictly between s_i and s_{i+1} has at least three distinct prime divisors by Theorem 1. Such an integer cannot be a semiprime. Hence $s_1, \ldots, s_r$ are consecutive in the sequence of all semiprimes.

Since t_i, t_j are distinct primes, for $i \neq j$ we have $\gcd(s_i, s_j) = p \gcd(t_i, t_j) = p$. Finally, $\Delta_i = p(h_{i+1} - h_i) = p\delta_i$. Multiplication by p preserves the strict inequalities and divisibility relations of the δ_i. The diameter equals $p(h_r - h_1) = ph_r$. All assertions follow. □

3 Infinitude of OEIS A131409

Corollary 1 (Infinitude of A131409)**.** *There are infinitely many triples of consecutive semiprimes $s_1 < s_2 < s_3$ with $\gcd(s_1, s_2, s_3) = 3$. Consequently, OEIS A131409 has infinitely many terms.*

Proof. OEIS A131409 lists the middle terms of triples of consecutive semiprimes with common prime factor 3 [3]. Apply Theorem 2 with $r = 3$ and $P = \{3\}$. It gives infinitely many triples

$$3q, \qquad 3(q + h_2), \qquad 3(q + h_3)$$

of consecutive squarefree semiprimes, with every pairwise gcd equal to 3. Their common gcd is therefore 3. The middle term $3(q + h_2)$ takes infinitely many distinct values as q ranges over the infinitely many qualifying integers. All these middle terms belong to A131409. □

Example. The first triple displayed in A131409 is

$$(3903, 3909, 3921) = 3(1301, 1303, 1307),$$

with middle term 3909 [3]. All three cofactors are prime, and the only semiprimes in $[3903, 3921]$ are the displayed three numbers.

4 Attribution, scope, and AI authorship disclosure

The analytic input is the established quantitative theorem in [1]. The use of congruences to force consecutive prime blocks, and increasing or divisible patterns of prime gaps, are treated in [2]. This note separates one OEIS consequence from the supplied combined proof. It does not determine first-occurrence values, establish novelty, claim publication priority, or represent an independent verification of the argument.

AI authorship and verification. Primary author (AI): ChatGPT (OpenAI). This standalone PDF was prepared from the supplied mathematical note with the shared argument retained and the OEIS-specific conclusion isolated. Its mathematical correctness has not been independently checked during this document-formatting task.

References

[1] J. Maynard, *Dense clusters of primes in subsets*, Compositio Mathematica **152** (2016), 1517–1554, Theorem 3.4. https://arxiv.org/abs/1405.2593.

[2] W. D. Banks, T. Freiberg, and C. L. Turnage–Butterbaugh, *Consecutive primes in tuples*, Acta Arithmetica **167** (2015), 261–266. https://arxiv.org/abs/1311.7003.

[3] Z. Seidov, *Middle of three consecutive semiprimes with common prime factor = 3*, The On-Line Encyclopedia of Integer Sequences, A131409 (2007). https://oeis.org/A131409.